Project Proposal

KrishiSathi

Integrated Agriculture & Livelihood Platform

Course	CSE-3200 — Software Development Project-II
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1 Introduction

1.1 Purpose

This document specifies the software requirements for **KrishiSathi**, an all-in-one mobile platform proposed to support Bangladeshi farmers, fish-farmers, and livestock keepers across the full agricultural lifecycle — from crop, pond, and livestock advisory through to direct market sale. It is intended as a shared reference for the course instructor and the project team, describing what the system does, whom it serves, and how it is structured.

1.2 Scope

KrishiSathi covers three primary livelihood domains — crop farming, fishery (pond and coastal shrimp), and livestock/poultry rearing — unified under one platform rather than treated as separate apps. The system provides location- and condition-based advisory, image-based disease detection, weather- and disaster-linked recommendations, a direct farmer-to-buyer marketplace, and a set of supporting services: labour matching, financial guidance, government scheme discovery, and offline-first access designed around Bangladesh's connectivity and literacy realities.

1.3 Problem Statement

- Farmers lack easy access to localized, condition-specific agricultural advice (soil, crop, weather).
- Middlemen (locally known as *dalal*) capture a large share of the profit margin that sits between farmer and end buyer.
- Crop, fish, and livestock diseases are often misdiagnosed or treated too late due to limited access to extension officers or veterinarians.
- Natural disasters — flood, cyclone, salinity intrusion, flash floods in haor basins — disproportionately affect farmer livelihoods, often with too little early warning to act on.
- Existing digital agri-services are fragmented: crop, fishery, and livestock advisory are rarely supported inside a single platform, and most are built around smartphone-only, high-bandwidth assumptions that do not match rural connectivity.
- Financial inclusion for small and marginal farmers remains low; many lack a documented credit history that formal lenders or subsidy programs can act on.

2 Overall Description

2.1 User Classes and Characteristics

Actor	Description
Farmer / Krishok	Uses advisory tools (crop, fish, poultry), uploads images for disease detection, lists produce for sale, hires labour, and tracks scheme eligibility.
Buyer / Trader	Browses listings, places bids or offers, and purchases produce directly from farmers, bypassing intermediary agents.

Actor	Description
Labourer	Lists availability and skills, gets matched with farmers needing seasonal work (harvest, pond, or khamar labour).
Admin / Krishi Officer	Verifies users, moderates price data and forum content, and responds to advisory escalations that automated tools cannot resolve.
Field Extension Worker (SAAO)	A Sub-Assistant Agriculture Officer or DAE-affiliated field worker who reviews escalated cases, validates disease-detection results in ambiguous cases, and can post verified local advisories.

2.2 Operating Environment

- Android/iOS mobile application (React Native + Expo), with an Android-first rollout given Android's dominant device share in rural Bangladesh.
- Cloud backend (Node.js/Express REST API) with PostgreSQL for structured data and MongoDB for unstructured content.
- Offline-first local storage (SQLite) with background synchronization, and a lightweight data mode that keeps sync payloads small on 2G/3G connections.
- A companion USSD/SMS channel for feature-phone users and for moments when the app itself cannot be reached (see § 4.9).

2.3 Design and Implementation Constraints Specific to Bangladesh

KrishiSathi's requirements are shaped as much by the operating context as by the farming domain itself. Three constraints recur across almost every module and are called out here rather than repeated in each section:

- **Connectivity is uneven.** Large parts of the target user base are on 2G/3G connections with intermittent coverage, so every feature needs an offline-tolerant fallback rather than treating connectivity as a given.
- **Literacy and language vary.** Bangla-first design with voice input/output is a baseline requirement, not an accessibility add-on, and UI copy should stay close to spoken register rather than formal/literary Bangla.
- **Trust has to be earned structurally.** Because the platform's core value proposition is disintermediating the middleman (*dalal*), farmer-buyer trust (payment, delivery, dispute handling) has to be designed for explicitly rather than assumed.

3 Functional Requirements — Modules

Module	Key Features	Priority
Crop Advisory	Soil- and location-based crop recommendation, disease detection via image (CNN), pest detection, fertilizer dosage calculator, weather-based irrigation advisory, yield prediction, seasonal crop calendar (Aus/Aman/Boro/Rabi), salinity- and flood-tolerant variety suggestions (BRRI dhan lines) for coastal and haor districts	**High**
Fishery Module	Pond water-quality based fish species suggestion, fish disease detection, feed calculator, harvest-time prediction, shrimp/gher salinity advisory, export-traceability logging for shrimp destined for processing plants	**High**
Livestock & Poultry	Breed advisory, vaccination schedule reminders, disease alerts (e.g. bird flu), feed-to-weight/milk-yield calculators	**Medium**
Weather & Disaster Advisory	Unified rain/flood/cyclone/heatwave alerts affecting crop, pond, and livestock decisions; BWDB flood-data and BMD weather integration; cyclone-shelter locator for coastal unions	**High**
Marketplace	Real-time market price feed, direct farmer-buyer matching (removes the middleman), multi-buyer bidding, price-trend forecasting, in-app MFS payment (bKash/Nagad/Rocket) with escrow-style holding on first transactions	**High**
Labour & Livelihood	Day-labourer marketplace for harvest/pond/khamar work; lean-season (Monga) alternative income suggestions for greater Rangpur and other historically affected districts	**Medium**
Financial & Govt Support	Cost-benefit calculator, microfinance/loan advisory (Bangladesh Bank refinance schemes, PKSF-linked NGOs), government subsidy tracker (BADC etc.), cold storage locator, basic digital credit-history logging for future loan applications	**Medium**
Support System	Bangla voice input/output, Krishi Batayon (16123) hotline integration, local officer (SAAO) directory, community forum, offline-first sync	**Low**
Feature-Phone & USSD Access	SMS/USSD advisory subscription for non-smartphone users; price alerts and weather warnings delivered as plain-text SMS	**Medium**
Gender-Inclusive Access	Voice-first flows and a simplified UI mode aimed at women farm managers, who run a substantial share of homestead and livestock operations but are statistically less likely to own or control a smartphone	**Medium**

4 Detailed Functional Requirements

4.1 Crop Advisory Module

- **FR-1.1** System shall recommend suitable crops based on user-entered location and soil type.
- **FR-1.2** System shall accept an uploaded leaf/plant image and return a probable disease name with confidence score.
- **FR-1.3** System shall suggest treatment (pesticide or organic remedy) for a detected disease.

- **FR-1.4** System shall calculate fertilizer dosage (kg/bigha, with katha and decimal unit toggles) from soil N-P-K input.
- **FR-1.5** System shall send irrigation/fertilizer advisory notifications based on weather forecast (e.g. suppress advice if rain is predicted).
- **FR-1.6** System shall display a seasonal crop calendar (Aus, Aman, Boro, Rabi) relevant to the user's district.
- **FR-1.7** System shall flag salinity- or flood-tolerant variety options (BRRI-released cultivars) when the user's district is tagged as coastal or haor-prone.

4.2 Fishery Module

- **FR-2.1** System shall suggest suitable fish species from user-entered pond water parameters (pH, ammonia, oxygen).
- **FR-2.2** System shall detect common fish diseases from an uploaded image or symptom description.
- **FR-2.3** System shall calculate feed quantity based on pond size and fish age/stock.
- **FR-2.4** System shall provide salinity-based advisory for coastal shrimp (chingri) farmers.
- **FR-2.5** System shall let shrimp farmers log basic batch information (stocking date, feed source, treatments applied) to support downstream export-traceability requirements, without requiring this for pond-only freshwater users.

4.3 Livestock & Poultry Module

- **FR-3.1** System shall provide breed selection guidance for poultry, dairy, and goat farming.
- **FR-3.2** System shall send vaccination schedule reminders per livestock type.
- **FR-3.3** System shall alert users to regional disease outbreaks (e.g. bird flu), sourced from Department of Livestock Services (DLS) advisories where available.

4.4 Weather & Disaster Advisory

- **FR-4.1** System shall fetch and display real-time weather data for the user's location, drawing on Bangladesh Meteorological Department (BMD) feeds where available, with a commercial API fallback.
- **FR-4.2** System shall issue flood/cyclone early warnings using BWDB and meteorological data sources.
- **FR-4.3** System shall tailor disaster alerts to the user's registered farming type (crop/pond/livestock).
- **FR-4.4** System shall show the nearest cyclone shelter and a basic evacuation note for users registered in coastal upazilas, sourced from local disaster-management office listings.

4.5 Marketplace Module

- **FR-5.1** System shall display current market prices for registered products, cross-checked against Department of Agricultural Marketing (DAM) data where available.
- **FR-5.2** System shall allow farmers to list produce for sale (quantity, price, location).
- **FR-5.3** System shall allow buyers to browse listings and submit offers/bids.
- **FR-5.4** System shall match farmers with the best available offer, bypassing intermediary agents.
- **FR-5.5** System shall forecast short-term price trends (e.g. seasonal demand spikes).
- **FR-5.6** System shall support in-app payment confirmation through mobile financial services (bKash, Nagad, Rocket), with an optional escrow-style hold released on delivery confirmation for a buyer-farmer pair's first few transactions.
- **FR-5.7** System shall let both farmer and buyer rate a completed transaction, building a lightweight reputation score that is visible on future listings.

4.6 Labour & Livelihood Module

- **FR-6.1** System shall allow labourers to register availability and skills.
- **FR-6.2** System shall match farmers' labour requests with nearby available labourers.
- **FR-6.3** System shall suggest alternative income sources during lean season (Monga), tuned to districts with a documented Monga history.

4.7 Financial & Government Support Module

- **FR-7.1** System shall calculate estimated cost, revenue, and profit margin per crop/fish/poultry cycle.
- **FR-7.2** System shall list applicable government subsidy schemes by crop/region.
- **FR-7.3** System shall display nearby cold storage facilities and their availability.
- **FR-7.4** System shall maintain a simple, farmer-visible transaction and yield history that can be exported as a summary document to support microfinance or agricultural loan applications.

4.8 Support System

- **FR-8.1** System shall support Bangla voice input and text-to-speech output.
- **FR-8.2** System shall provide access to the Krishi Batayon (16123) hotline and local officer directory.
- **FR-8.3** System shall function in offline mode with local caching and later synchronization.
- **FR-8.4** System shall provide a community forum for farmer-to-farmer discussion.

4.9 Feature-Phone and Low-Bandwidth Access

- **FR-9.1** System shall allow a user to subscribe to price and weather alerts over plain SMS using a short registration code, without requiring the mobile app.
- **FR-9.2** System shall expose a USSD menu for checking today's market price of a small set of registered crops.
- **FR-9.3** System shall degrade gracefully on the mobile app itself: image upload and forum browsing may be deferred on a detected low-bandwidth connection, while text-based advisory and alerts remain available.

5 Non-Functional Requirements

Category	Requirement
Usability	Bangla-first UI, voice input/output for low-literacy users; UI text kept at a spoken-register reading level rather than formal Bangla.
Availability	Offline-first design with local SQLite cache and background sync; core advisory content pre-cached per district on first login.
Performance	Disease-detection inference response under 5 seconds on average mobile hardware; SMS/USSD responses under 10 seconds.
Scalability	Backend should support horizontal scaling as the user base grows across districts, with district-level data partitioning anticipated ahead of nationwide rollout.
Security	Farmer and buyer data encrypted in transit (HTTPS) and at rest for financial and identity fields; role-based access control; MFS payment flows never store raw account credentials, only tokenized references.
Reliability	Weather/flood alerts must have redundant data sources to avoid missed warnings.
Accessibility	Voice-first navigation path usable without reading UI labels; large-touch-target mode for older users.
Inclusivity	Interfaces and onboarding flows are tested with women farm managers specifically, not assumed to transfer automatically from male-farmer usability testing.

6 System Architecture Overview

KrishiSathi follows a three-tier architecture, with a fourth channel layer added specifically to reach users the mobile-app tiers alone would miss.

- **Presentation Layer:** React Native + Expo + TypeScript mobile client (Android/iOS), offline-capable via local SQLite.
- **Application Layer:** Node.js/Express REST API handling business logic, authentication, and orchestration of ML inference services.
- **Data & ML Layer:** PostgreSQL (structured data — users, listings, prices), MongoDB

(unstructured — images, forum posts), Python-based ML services (scikit-learn for recommendation, TensorFlow/PyTorch CNN for disease detection) served via a dedicated inference API.

- **Alternate Access Layer:** An SMS/USSD gateway sitting in front of the same application layer, so alerts and price look-ups reach feature-phone users through the identical backend logic rather than a parallel system.

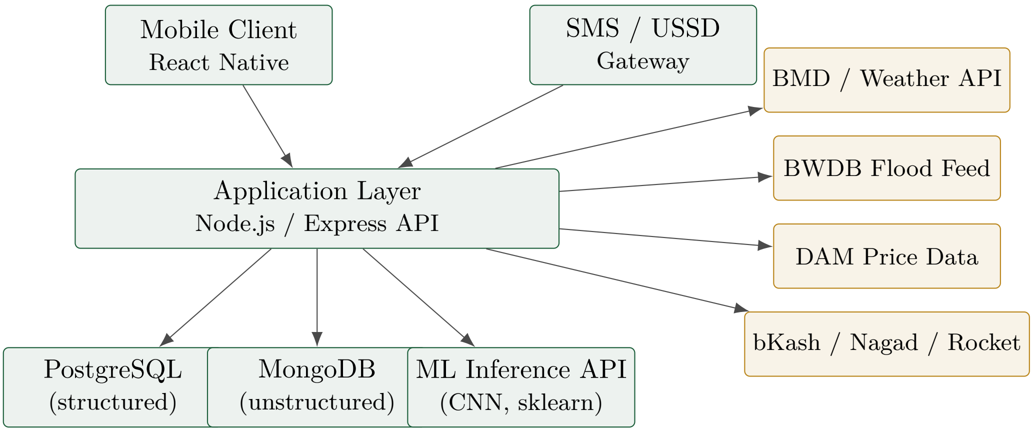

Figure 1: High-level component view of KrishiSathi's backend and its external integrations.

External integrations include a weather API (BMD, with OpenWeatherMap as fallback), the BWDB flood-data feed, Department of Agricultural Marketing (DAM) price data, Firebase push notifications, mobile financial service APIs (bKash/Nagad/Rocket) for the marketplace, and an SMS gateway fallback for low-connectivity users.

7 Assumptions and Constraints

- Initial deployment will target a limited set of pilot districts, chosen to cover at least one coastal, one haor, and one Monga-prone district, before nationwide scale-up.
- ML models will initially be trained on publicly available datasets (e.g. PlantVillage) and refined with locally collected data over time; accuracy in the pilot districts is expected to lag general benchmarks until local data accumulates.
- Real-time government data sources (BWDB, DAM, BMD) may have limited API availability; fallback to periodically updated cached data is assumed.
- MFS integration depends on each provider's merchant-API onboarding process, which may gate the escrow-style payment feature's release timeline independently of the rest of the marketplace module.

8 Scope Evolution and Design Decisions

KrishiSathi began as a broad, almost all-encompassing idea — one platform for every farming domain, every disaster type, and every underserved user segment in rural Bangladesh. The requirements above already reflect a first pass of narrowing that down to something a small team can realistically build and evaluate within the course timeline. As we continued researching the target users, a few features and goals were consciously prioritized, and a few were set aside for later phases. This section records that reasoning so it remains visible for future design and implementation decisions, rather than presenting the current scope as final or fully settled.

8.1 What we chose to add

- **Mobile financial service (MFS) payment support.** The original marketplace design assumed listings and matching were enough to solve the middleman problem. Talking through realistic transaction flows made it clear that removing the middleman without addressing payment and delivery trust just recreates the same trust gap in a different place — so escrow-style MFS payment and a lightweight reputation score were added rather than left as a ”future work” item.

- **SMS/USSD access.** A smartphone-and-data-plan assumption would have quietly excluded a meaningful share of the target users. Rather than treat this as an accessibility afterthought, it is now a first-class access channel sharing the same backend.

- **Region-specific disaster and variety guidance** (haor flash floods, coastal salinity, cyclone shelters, Monga-season income support). Bangladesh's agricultural risk is not uniform across the country, and a generic ”weather alert” feature undersold what the platform could realistically do for the districts that need it most.

- **Gender-inclusive access considerations.** Women manage a substantial share of homestead crop, poultry, and livestock work in rural Bangladesh but are less likely to own or control the household smartphone. Voice-first flows and a simplified UI mode were added specifically with this usage pattern in mind, rather than assuming general usability testing would cover it.

8.2 What we chose to defer or drop

- **Full yield-prediction modeling for all crop types** was scaled back for the initial release. Reliable yield prediction needs multi-season local data that a pilot simply will not have yet; the module now ships with a narrower, better-supported set of crops and is explicit that yield figures are indicative rather than authoritative until more data is collected.

- **Automated dispute resolution** for marketplace transactions was considered and set aside in favor of routing disputes to a human admin/Krishi officer for now. An automated arbitration flow is a reasonable long-term goal, but building it before there is real transaction volume to learn from risked over-engineering a feature nobody would trust yet.

- **A general-purpose agri-chatbot** covering open-ended farming questions was dropped from this phase. An open-ended question-answering surface is considerably harder to keep accurate than the structured advisory flows already in scope, and an incorrect answer in this domain carries real consequences for a farmer's livelihood. Structured advisory with human escalation therefore remains the safer default until reliable verification tooling for open-ended answers is in place.

- **IoT sensor integration** (soil moisture, pond water quality) was moved to future scope rather than this SRS. It depends on hardware distribution and farmer-side maintenance that are outside what a software team can control within this course project, even though it remains a natural extension once the software side is proven.

- **Nationwide launch at day one** was replaced with a staged, pilot-district rollout. Trying to be everything for every district immediately would have diluted both the data quality (per-district crop calendars, disease patterns, price feeds) and the team's ability to actually support users during early failures.

8.3 How this list is expected to change

This section is expected to be revisited as the pilot districts are chosen and early user feedback comes in; some deferred items may move back into active scope once their prerequisites (data, partnerships, or provider onboarding) are in place, and some current requirements may in turn be trimmed if they prove to add more complexity than value in practice.

9 Future Scope

- Integration of IoT soil-moisture and pond sensors for automated advisory.
- Expansion of ML models to cover a wider range of crop, fish, and livestock diseases, and a broader set of BRRI/BARI-released varieties.
- Formal microfinance/insurance partnership integration, building on the transaction and yield history already captured in § 4.7.
- Automated, data-backed dispute resolution for the marketplace once transaction volume supports it.
- A carefully scoped, human-reviewed advisory chatbot once verification tooling exists to keep open-ended answers reliable.